3–3 TROUBLESHOOTING

3–3–1 Prestartup

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK DISCONNECT POWER FROM COMPACT PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock, and tag.
2. Open outer door to Compact PDU.
3. Turn main circuit breaker OFF.
4. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

Note

The following step applies to models 317099 P8 through P16 except P12. Adjust all three logic boards (A14A2, A14A3, A14A4). See Illustration 3–16.

5. On each logic board, attach ohmmeter leads to u9 pin 6 to U0 pin 8, if the logic board is oriented as shown in Illustration 3–16. Adjust R33 for ohmmeter to read 105K ohms.
6. Check that all plug connectors on circuit boards are tight. Refer to Section 3–6–4, PM Procedure, Step 17.
7. Models 46–317099 P8 through P16 except P12 only: Tighten screws on all 18 SCRs (54 screws; 3 per SCR). Torque to 30 lb in (34 Nm). See Illustration 3–17.
8. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center right edge of door.
9. Unlock facility circuit breaker and turn ON.

Note

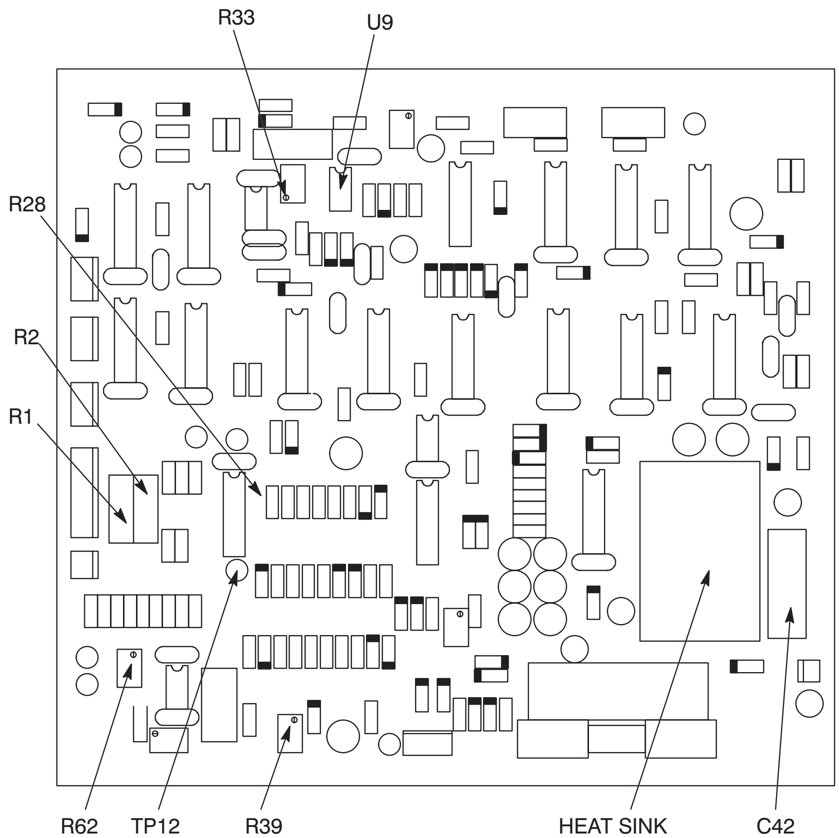
There is a 100:1 voltage stepdown at test points. For example, 480 VAC in Compact PDU is measured as 4.8 VAC at test points.

10. Models 46–317099 P8 through P16 except P12 only: Measure facility line–to–line voltage at input test points. Refer to Table 3–3.

TABLE 3–3
VOLTAGE INPUT BY MODEL

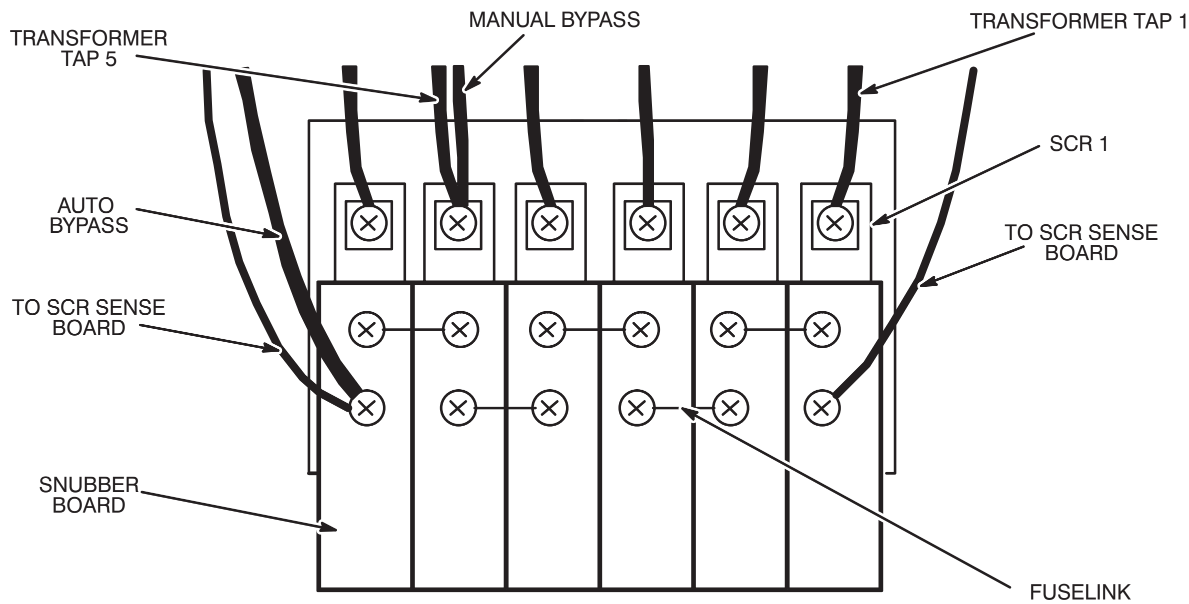
COMPACT PDU PART NO.	NOMINAL INPUT LINE–TO–LINE	INPUT AT INPUT TEST POINTS
46–317099P8/P10/P15/P16	480 V 60 Hz regulated, filtered	4.12 to 5.02 V
46–317099P9/P11/P13/P14	380 V, 415, or 480 V 50/60 Hz regulated, filtered	3.27 to 3.97 V or 3.57 to 4.34 V

3−3−1 Prestartup (continued)



LOGIC BOARD
ILLUSTRATION 3−16

3−3−1 Prestartup (continued)



SCR BANK
ILLUSTRATION 3−17

3−3−2 Basic Check

1. Unlock facility circuit breaker and turn ON.
2. Models 46−317099P12: go directly to Step 6.
3. Turn bypass switch to AUTO position.
4. Turn main circuit breaker ON. Press emer. stop reset switch.

Note

After turning main circuit breaker ON there is a 6−second built−in delay before power is applied to SCRs. Detect an SCR failure by measuring neutral−to−line output voltage of each phase. Test point voltages are in the range 0−5 VAC.

5. Set meter to 2 VAC scale.

Note

Measured voltage should be close to zero. If measured voltage is greater than 0.2 VAC, there is a shorted SCR in regulation panel. Refer to Section 3−3−4, SCR troubleshooting.

a. Observe meter after 6 seconds. Verify test point A line−to−neutral output voltage of 1.20 $\pm$ 0.03 V.

3–3–2 Basic Check (continued)

b. Press power off switch.

c. Turn main circuit breaker OFF then ON to reset.

d. Move meter lead to output test point B. Perform Steps a. through c.

e. Move meter lead to output test point C. Perform Steps a. through c.

6. Turn main circuit breaker ON.

7. Press Emer. Stop Reset switch.

8. Turn load panel circuit breakers ON.

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN COMPACT PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

9. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

10. Check each set of output line–to–neutral voltages at TB10 (A2). Refer to Illustration 3–6, 3–7, 3–8, and 3–9.

TABLE 3–4
VOLTAGE INPUT BY MODEL

COMPACT PDU PART NO.	NOMINAL INPUT LINE–TO–LINE	OUTPUT LINE–TO–NEUTRAL
46–317099P8/P10/P15/P16	480 V 60 Hz regulated, filtered	120 ± 3 V
46–317099P9/P11/P13/P14	380 V or 415 V 50/60 Hz regulated, filtered	120 ± 3 V
46–317099P12	380 V, 415, or 480 V 50/60 Hz unregulated	120 ± 3 V

Models 46–317099P12: Basic check is complete. Continue with Step 14. Models 46–317099 P8 through P16 except P12: Continue with Step 11.

11. Adjust logic board for phase A as follows. See Illustration 3–16.

a. Measure AC voltage across resistor R1 or R2. Verify that voltage is between 10–100mv.

3–3–2 Basic Check (continued)

Note

If voltage is zero, (1) there is no load, or (2) current sensor is defective or disconnected. Logic board does not regulate if voltage is zero.

b. Measure DC voltage across C42. Adjust R62 so voltage measures 8.5–9.0 VDC.

c. Adjust R39 counterclockwise so that LED 6 on driver board is lit.

d. Measure DC voltage across C42. Adjust R62 so voltage measures 8.5–9.0 VDC.

e. Monitor neutral–to–phase–A output voltage. Adjust R39 so output line–to–neutral voltage is 1.20 $\pm$ 0.03 V.

12. Adjust logic boards for phase B and phase C. Refer to Step 11.

13. Verify that voltage at TP12 to ground on each logic board is 10 $\pm$ 0.5 VDC. If voltage is not within these limits, replace logic board.

14. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center of right edge of door.

15. Close outer door to Compact PDU.

3–3–3 Functional Check

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK DISCONNECT POWER FROM COMPACT PDU AND LOCK OFF BEFORE YOU PERFORM THE FOLLOWING PROCEDURE.

1. Turn facility circuit breaker OFF, lock, and tag.

2. Check and replace any fuselinks which are discolored or which are not continuous.

3. See Illustration 3–17. For each SCR bank:

Note

SCRs are numbered from right to left. SCR number six is on the left when SCRs are oriented as shown in Illustration 3–17.

a. Turn bypass switch to the manual position.

b. Verify that resistance between an SCR transformer lead and a snubber board solder pad exceeds 1K ohms. If not, refer to Section 3–3–4, SCR Troubleshooting.

3−3−3 Functional Check (continued)

c. Turn bypass switch to AUTO position.

4. Check snubber boards for discoloration and rupture of foil.
5. Check that all snubber board plug connections are tight.
6. Check that logic, driver, and SCR sense boards are firmly supported on standoffs.
7. Check that all plug connectors on circuit boards are tight. Refer to Section 3−6−4, PM Procedure, Step 17, for list of connectors.
8. Check for heat discoloration near heat sink area on logic boards. See Illustration 3−16.
9. Check that all connectors to input/output filter capacitors are tight.
10. Unlock facility circuit breaker and turn ON.
11. Turn main circuit breaker ON.
12. Press emer. stop reset switch.

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN COMPACT PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

13. Press power off switch. This shunt trips main circuit breaker.
14. Remove fuselink from any one SCR bank.
15. Turn main circuit breaker OFF then ON to reset. Main circuit breaker will trip in about 10 seconds. If it does not trip, replace SCR sense board.
16. Install fuselink. Turn main circuit breaker OFF then ON to reset.
17. Press emer. stop reset switch.
18. Check output line−to−neutral voltage at test points. Verify voltage of 1.20 $\pm$ 0.03 V. If voltage is out of specification, refer to Section 3−3−2, Basic Check, for logic board adjustment.
19. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4−20 screw in center right edge of door.
20. Turn load panel circuit breakers ON.
21. Close outer door.

3−3−4 SCR Troubleshooting (Models 46−317099 P8 through P16 except P12)

1. Turn main circuit breaker OFF.

Note

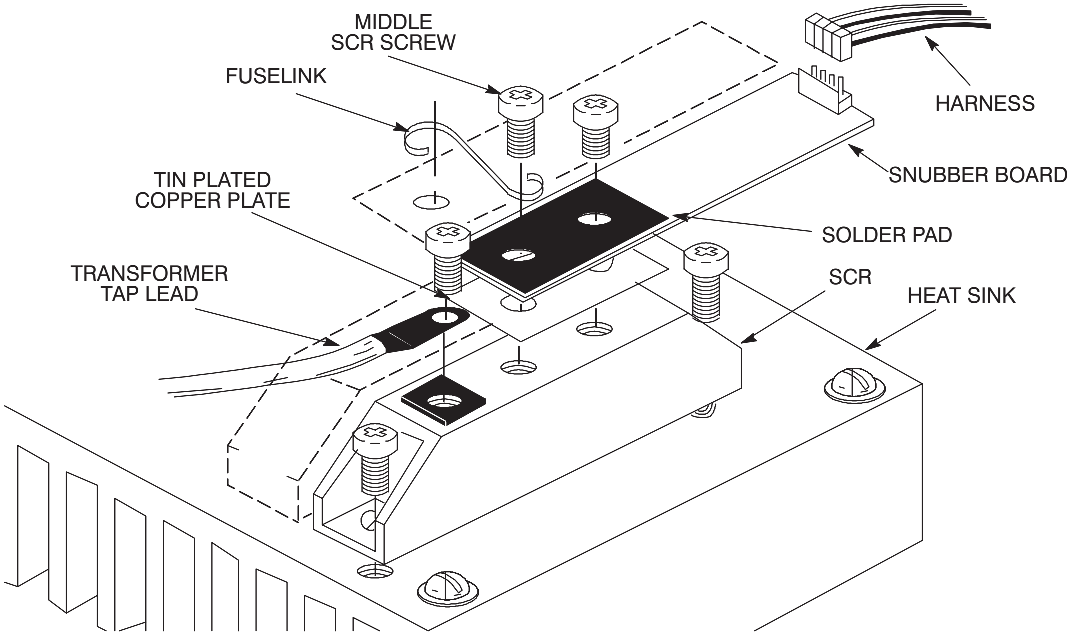
SCRs are numbered from right to left. SCR number six is on the left when SCRs are oriented as shown in Illustration 3−17.

2. Turn bypass switch to the manual position.

3. Disconnect fuselinks between snubber boards. Do not remove snubber boards.

4. For SCR number one:

a. Measure resistance between transformer tap connection and solder pad area of snubber board. If reading is greater than 1000 ohms, proceed to Step 5. If reading is less than 1000 ohms, continue with this step. Either SCR, snubber board, or driver board is defective. See Illustration 3−18.



SCR ASSEMBLY
ILLUSTRATION 3−18

b. Disconnect harness between driver board and snubber board. Measure resistance between transformer tap lead and each middle SCR screw.

c. If reading is greater than 1000 ohms then driver board is defective. Replace driver board. Proceed to Step 4f. If reading is less than 1000 ohms then either SCR or snubber board is defective. Continue with step 4d.

d. Disconnect snubber board from SCR. Measure resistance between tap connection and screw in middle of SCR. If resistance is less than 1000 ohms, replace SCR. If resistance is greater than 1000 ohms, replace snubber board.

3−3−4 SCR Troubleshooting (Models 46−317099 P8 through P16 except P12) (continued)

   e. Reconnect snubber board to SCR.

   f. Reconnect harness between driver board and snubber board.

5. Repeat Step 4 for each remaining SCR in the SCR bank.

6. Reconnect fuselinks between snubber boards. Replace any fuselink which is open or discolored.

7. Replace any snubber board which has cracked or discolored foil pads on either side of board.

8. Repeat Steps 2−8 for each remaining SCR bank.

9. Turn bypass switch to AUTO position.

10.

Note
Steps 11a, 11b and 11c must be finished within six seconds.

   a. Turn main circuit breaker ON.

   b. Measure voltage across an SCR between transformer tap connection and foil area of snubber board.

   c. Turn main circuit breaker OFF.
      If measurement is $\geq$ 2 V, perform Step 11 for next SCR.
      If measurement is less than 2 V go to Step 12.
      If all SCRs have voltage measurements $\geq$ 2 V, go to Step 13.

11. SCR or snubber board is defective. Perform this step to determine which part is defective.

   a. Disconnect snubber board from SCR.

   b. Measure resistance between tap connection and screw in middle of SCR.

   c. If resistance is less than 1000 ohms, replace SCR. Refer to Section 3−4−2, Replace an SCR.
      If resistance is $\geq$ 1000 ohms, replace snubber board. Refer to Section 3−4−1, Service a Circuit Board.

   d. Continue with Step 10.

12. Turn main circuit breaker ON.

3−3−5 Filter Troubleshooting

1. Open outer door of Compact PDU.

2. Turn main circuit breaker OFF.

3. Turn all load panel circuit breakers OFF.

3–3–5 Filter Troubleshooting (Models 46–317099 P8 through P16 except P12) (continued)

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN COMPACT PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

4. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4–20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.

5. Turn main circuit breaker ON. If main circuit breaker shunt trips, check accuvar "O.K." light is ON. If fail light is ON replace accuvar.

   a. Verify that the green OK light is ON.

   Note

   Voltage measured at line side will be nominal input volts to the PDU. These are the terminal lugs in the front of the filter assembly

   b. Verify that line–to–line voltage on load side of filter terminal block at rear of the filter assembly is in operating voltage range (208Y/120).

6. Check for visible damage to filter.

   a. If a terminal block is damaged, replace terminal block.

   b. If surge suppressor is damaged, this will be indicated by the red fail light ON. Replace surge suppressor. Refer to Section 3–4–5, Replace Surge Suppressor.

   c. If any capacitors damaged, replace the capacitors.

7. If there is no visible damage, disconnect wires from line or load side of surge suppressor.

8. Turn main circuit breaker.

   a. If main circuit breaker shunt trips, the problem is in the PDU

   b. Replace the power wires to the line filter.

9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4–20 screw in center of right edge of door.

10. Turn main circuit breaker ON.

11. Turn load panel circuit breakers ON.

3−3−5 Filter Troubleshooting (Models 46−317099 P8 through P16 except P12) (continued)

12. Close outer door.

3−3−6 Miscellaneous Problems

PROBLEM: Main circuit breaker fails to shunt trip.

PROCEDURE:

DANGER!!

FATAL SHOCK HAZARD!! LETHAL VOLTAGES EXIST WITHIN COMPACT PDU DURING THIS CHECK. FOLLOW THE STEPS BELOW EXACTLY. FAILURE TO DO SO COULD RESULT IN SEVERE INJURY OR DEATH.

1. Disconnect connector on left side of main circuit breaker.
2. Measure shunt trip coil resistance with circuit breaker turned ON. It must read <10 ohms. With circuit breaker OFF, the shunt trip coil must be open. If either condition is not met, replace circuit breaker.
3. Reconnect connector behind main circuit breaker.
4. Turn main circuit breaker ON.
5. Measure shunt trip coil voltage on cable side of unplugged connector J1 with power off switch pressed. If voltage is 24−28 VDC, replace main circuit breaker. Refer to Section 3−4−3, Replace Main Circuit Breaker.
6. If no voltage, check fuses F1 and F8 on A8. If fuse failed, replace fuse.

PROBLEM: Main circuit breaker shunt trips on powerup with bypass switch in AUTO position (Models 46−317099 P8 through P16 except P12).

PROCEDURE:

1. If SCR failure is lighted, replace any open fuselinks, defective snubber boards, or shorted SCRs. Press fail reset switch to clear SCR failure light.
2. Turn main circuit breaker OFF. Disconnect SCR sense board J9. Turn main circuit breaker ON. If breaker does not shunt trip, replace SCR sense board.

PROBLEM: No LED on driver board is lit, except LED 7 (Power OK), with bypass switch in AUTO position (Models 46−317099 P8 through P16 except P12).

PROCEDURE:

1. Check connection between logic board and driver board.

3−3−6 Miscellaneous Problems (continued)

2. Check input power for clockwise phase rotation.
3. Measure voltage across SCR between transformer tap lead and snubber solder pad. When SCR is conducting, measured voltage should be less than 1.2 VAC.
4. If no SCRs are conducting, replace driver board. Refer to Section 3−4−1, Service a Circuit Board.
5. If SCRs are still not conducting, replace logic board. Refer to Section 3−4−1, Service a Circuit Board.
6. Check voltage across R1 or R2 on logic board. Voltage is typically 10 − 100 mVAC. If it is zero, current signal is disabled from logic board. Check current sense transformer connections. Repair any opens.

PROBLEM: All LEDs on driver board are off.

PROCEDURE:

1. Check interconnect cables to driver board.
2. Check LED 1 on logic board (Power OK). If LED 1 is not lighted, replace logic board.

PROBLEM: No voltage output in AUTO mode (Models 46−317099 P8 through P16 except P12).

PROCEDURE:

1. Check that main input breaker is ON.
2. Check that load panel circuit breakers are ON.
3. Check that facility circuit breaker is ON.
4. Check that line input power connections are made to H_1, H_2, and H_3 transformer terminal lugs. See Illustration 3−14.

PROBLEM: Fuselink opens during powerup (Models 46−317099 P8 through P16 except P12).

PROCEDURE:

1. Check that primary RC filter is installed and connected.
2. Check that capacitor connections are tight.
3. Check adjustment for R33. Refer to Section 3−3−1, Prestartup.

PROBLEM: Main circuit breaker trips.

PROCEDURE:

1. Perform prestartup checks. Refer to Section 3−3−1, Prestartup.
2. Models 46−317099 P8 through P16 except P12: If problem persists, replace driver and logic boards. Refer to Section 3−4−1, Service a Circuit Board.

3−3−6 Miscellaneous Problems (continued)

PROBLEM: Compact PDU does not regulate (Models 46−317099 P8 through P16 except P12).

PROCEDURE:

1. Perform prestartup. Refer to Section 3−3−1, Prestartup.
2. Perform basic check. Refer to Section 3−3−2, Basic Check.
3. If problem persists, replace driver and logic boards. Refer to Sections 3−4−1, Service a Circuit Board.

PROBLEM: High line−to−neutral output voltage measured within six seconds after powerup (Models 46−317099 P8 through P16 except P12).

PROCEDURE: There is a shorted SCR. Refer to Section 3−3−4, SCR Troubleshooting.

PROBLEM: Regulation panel is running hot.

PROCEDURE:

1. Check for loose power connections.
2. Check that fans are operating.
3. Check for loose connections between driver boards and snubber boards.
4. Check for loose connections between snubber board and SCR.
5. Replace snubber board for conducting SCR if problem persists. Refer to Section 3−4−1, Service a Circuit Board.

PROBLEM: Audible noise comes from regulation panel.

PROCEDURE:

1. Verify that voltage across conducting SCR is less than 1.2 VAC.
2. Perform prestartup. Refer to Section 3−3−1, Prestartup.
3. Perform basic check. Refer to Section 3−3−2, Basic Check.
4. Replace driver boards if problem persists. Refer to Section 3−4−1, Service a Circuit Board.

PROBLEM: Gradient, RF, and Verify Cabinet circuit breakers trip without a power off actuation.

PROCEDURE:

1. Verify that input voltage is present. Facility input power may have failed.
2. Verify main circuit breaker is ON.
3. Press emer. stop reset, and manually reset circuit breakers.